Basis

https://en.wikipedia.org/wiki/Basis_(linear_algebra)

Let V be a vector space.
A basis of a vector space can be **constructed**:

Any vector v1
Any vector v2 $\notin$ span {v1}
Any vector v3 $\notin$ span {v1, v2}
...

Basis B = {v1, ..., vn}
span B = V

Similarly, given any set of vectors B0, it can be extended to a basis:

Any vector v1 $\notin$ span B0
Any vector v2 $\notin$ span B0 $\cup$ {v1}
....

Basis B0 $\cup$ {v1,...,vk}

Dimension

The size of a basis is called the **dimension of V**

dim V = |B| any basis B

Dimension is well-defined since all bases must have the **same size**:
Indeed, if {wi} is another basis with more vectors, let

$$w_j = \sum_{i<n} a_{ij} v_i \in \text{span } B$$

[wj] = vector (a1j,...,anj) coefficients of wj

A = matrix { [w1] ... [wn] } = { aij } coefficients of w1...wn

A is invertible since w1...wn are linear independent.
Hence, by inverting the coefficients of wn+1

[yi] = A^-1 [wn+1]

$$\Rightarrow w_{n+1} = \sum_{i<n} y_i w_i$$

i.e. wn+1 as a linear combination of w1...wn

which is a contradiction.

Matrix multiplication

Matrix multiplication has two **dual** interpretations:

- **column aj** ∈ **F^m** vs **row vector Ai** ∈ **F^n**
- **linear combination** ∑ **xj aj** vs **application Ai*(x)** of dual Ai* ∈ F^n*

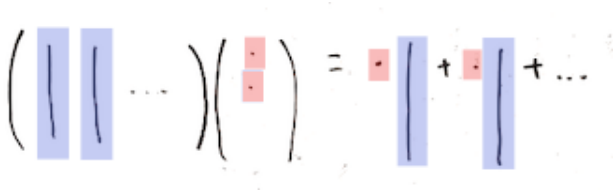
Let A = {aij} be an m x n matrix in field F.

aj = column (a1j, a2j, ..., anj) j = 1,..., m
Am = row (ai1 ai2 ... aim) i = 1,..., n

Ax x ∈ F^m
= { aij } x

Column vectors F^m
= (a1 a2 ...) x = x1 a1 + x2 a2 + ... **linear combination** of **column vectors aj** ∈ **F^m**

= ∑j xj aj

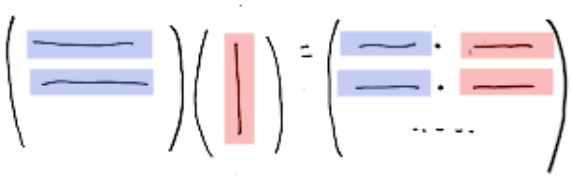


Rows vectors F^n*
= | a1 | x = | A1*(x) | **application** of **row vectors Ai*** ∈ **F^n***
| a2 | | A2*(x) |
...
....

Ai*: F^n → F
Ai*(x) = Ai ● x

= { Ai*(x) }i commutes
= { x*(Ai) }i

= { Ai ● x } dot product
= { x ● Ai }



Let b={vi} be a basis of V.
The **coefficients [v]b** of v in basis e

v = ∑i ki vi linear combination

[v]b = { ki }i coefficients of v
= (k1 ... kn)

If V is finite, then

v = ∑i ki vi

= B [v]b where B=(b1...bn) as column matrix

[v]b = B^-1 v

$$v = \sum_i k_i\ v_i$$

$$= (v_1\ v_2 \ldots) \begin{pmatrix} k_1 \\ k_2 \\ \vdots \end{pmatrix}$$

Let b' = {v'i} be another basis with matrix B'.
Then

[v]b' = B'^-1 v = (B'^-1 B) B^-1 v
= (B'^-1 B) [v]b

[v]b' = B^'-1 v = B^-1 (B B'^-1) v
= [B B'^-1v]b

B' = (B' B^-1) B
vi' = (B' B^-1) vi

Hence

[v]b' = A[v]b change of **coefficients** A b -> b'

[v]b' = [Cv]b change of **vector** C b -> b'

B' = C^-1 B **basis transform** B^-1 b -> b'
vi' = C^-1 vi

where

A = B'^-1 B
C = B B'^-1

Projection
https://en.wikipedia.org/wiki/Projection_(linear_algebra)

Let P: V → V be a linear map with P^2 = P.
Then any v ∈ V can be **uniquely decomposed** into Im P and Ker P

v = **Pv** + **v - Pv** note: Ker P since P(v - Pv) = Pv - Pv = 0
∈ Im P ∈ Ker P

This is unique since if

u + w = u' + w' some u, u ∈ Im P, w, w' ∈ Ker P
=> 0 = P(u-u') + P(w-w') = u-u' + 0
=> u = u'

Hence

V = Im P ⊕ Ker P

and P is called a **projection of V into Im P.**

Conversely, given any subspace W, there is a unique **projection P: V → V into W**
Let v1...vk be a basis of W, and **extend** it to a **basis v1...vn** of V. Let

P: V → V
Pvi = vi if i=1...k
0 o.w.

Then P is a projection of W

P = P^2
Im P = span {v1...vk}
= W

Ker P = span {vk+1...vn}

Projections are NOT unique for a given subspace.
Indeed, we can just choose a different **basis extension** {vk+1...vn} of W
See example below

Complementary projection
Projections have a **dual nature**.
Let

Q: V → V
Q = I - P

Then

QQv = Q(v - Pv) for any $v \in V$
= (v - Pv) - P(v - Pv)
= v - Pv
= Qv

=> Q^2 = Q

Hence Q is a projection.

Qv = (v - Pv) for any $v \in$ Ker P
= v

conversely

P(Qv) = Pv - P^2v
= v
=> $Qv \in$ Ker P

Hence

Im Q = Ker P

Q is a **projection into Im Q = Ker P.**

Overall

Im P = Ker Q
Ker Q = Im P

V = Im P ⊕ Ker P

= Ker Q ⊕ Im Q

***) Example:** different projections of the same space.
Let V = R^2 and W = x axis = span (1,0)
Define projections

P: R^2 → R^2	**P': R^2 → R^2**
P(1,0) = (1,0)	P'(1,0) = (1,0)
P(0,1) = 0	P'(1,1) = 0

Then P and P' are both projections of the x-axis but

$P \neq P'$

Indeed, any (x,y) has unique decomposition

(x,y) = (x,0) + (0,y)
∈ Im P ∈ Ker P

(x,y) = (x-y, 0) + (y,y)
∈ Im P' ∈ Ker P'

Theorem: Basic properties (Projection)

Eigenvalues The eigenvalues are precisely all 1

eigenvalue	1
eigenvector	v ∈ Im P
multiplicity	dim Im P

Hence

tr P = ∑ 1 over basis Im P
= dim (Im P)

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== Eigenvalue ==

eigenvalue, determinant, trace
The significance is that all are **invariant**s.
To be precise, they are all **invariants** under **basis** choice.

Eigenvalues:

value Avi = λivi i = 1...n

change of basis A'vi' = λi vi'
where E = change of basis
A' = EAE^-1
vi' = Evi

Both trace and determinant are forms of **canonical invariants** derived from

eigenvalues.
Trace is a **canonical additive** invariant.
Determinant is a **canonical multiplicative** invariant.

trace	tr A = $\sum \lambda$i
determinant	det A = $\prod \lambda$i

trace can be seen as a form of **matrix sum**.
determinant can be seen as a form of **matrix product**.

	scalar	~	matrix
<=>	eigenvalue		

sum	c (a+b) = ca + cb	tr c(A+B)	= c tr(A) + c tr(B)
	$\sum c(\lambda i + \eta i) = c\sum\lambda i + c\sum\eta i$		
			= tr c(B + A)
	$= c\sum(\eta i + \lambda i)$		
product	ab = ba	det (AB)	= det(A) det(B)
	$\prod \lambda i\eta i = \prod\lambda i \prod \eta i$		
			= det (BA)
	$= \prod\eta i\, \lambda i$		
	ca	det (cA)	= a^n det(A)
	$\prod c\lambda i = c^n \prod \lambda i$		

trace and determinant are canonical in that these properties are **unique characterizations**

trace: if f: GLn → F where

linear	f(c(A+B))	= c f(A) + c f(B)
identity	f(I)	= n
semi-commute	f(AB)	= f(BA)

then f = tr

det: if f: GLn → F where

multiplicative	f(AB)	= f(A) f(B)	
identity	f(I)	= 1	
diagonal	f({aii})	= $\prod$ aii	where a diagonal matrix {aii}

then f = det

Theorem: Basic properties

Basic

canon	tr A	= $\sum \lambda$i
	det A	= $\prod \lambda$i

A^k	A^k has the same eigenvectors as A

A^k v = λ^k v

eigenvalue λ^k
eigenvector v

Inverse A is invertible if and only if all eigenvalues are non-zero

A^-1 has the same eigenvectors as A

A^-1 v = 1/ λ v

eigenvalue 1/λ
eigenvector v

Unitary If A is unitary, then every eigenvalue is a unit

$|\lambda| = 1$ since $||v|| = ||Av|| = |\lambda| \, ||v||$
$\Rightarrow |\lambda| = 1$

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== Dual vector space V* ==

https://en.wikipedia.org/wiki/Dual_space

Linear functional
Dual (vector) space V*
A **linear functional** of a F-vectors space V is a **F-linear map f** on V

f: V → F

linearity
f(kv) = kf(v) scalar multiple
f(kv + vw) = f(v) + f(w) addition

The **dual space V*** is the set of all linear functionals of V.
V* is a **vector space**.

zero	0 ∈ V*	
scalar multiple	kf ∈ V*	any f ∈ V*, k ∈ F
addition	f + g ∈ V*	any f, g ∈ V*

Interpretation: coefficients v*(vi)
Let b={vi} be a basis of V.
Let v = ∑ ki vi, w = ∑i ci vi ∈ V. Let

[v]b = { ki }i vector of coefficients of v under basis E

= (k1 ... kn) if finite

hence the usual dot product

[v]b ● [w]b = ∑i ki ci

Let v = ∑ ki vi ∈ V. Then the **coefficients v*** of v are a linear functional

v*: V → F
v*(vi) = ki

v*(w) = ∑ci v*(vi) any w = ∑i ci vi
= ∑i ci ki

= [w]b ● [v]b

Conversely, let f: V → F be any linear functional.
Then f is determined by the values on the **basis { f(vi) }.**

f: V → F
f(vi) ∈ **F** each basis vector vi

f(w) = ∑i ci f(vi) any w = ∑i ci vi ∈ V

= [w]b ● { f(vi) }

Hence if V is **finite dimensional**, then f is exactly a coefficient v*

f = v* where v = ∑i f(vi) vi

Hence V* also has a **dual basis vi*: V → F** and
V and V* are **isomorphic**.

basis vi*
vi*: V → F
vi*(ej) = δi,j note: Kronecker delta
δi,i = 1 and δi,j = 0 otherwise
= [vi]b ● [vj]b

isomorphism V ≃ V*
V ≃ V*
V → V*
v ↦ v*
dim V = dim V*

In summary:

	V*		**V**
dimension	dim V*	=	dim V
basis	vi*: V → F	<=> vi*(vi) = δi,j	vi ∈ V
element	**coefficients** v*		**vector v**
	v*: V → F	<=>	v ∈ V
	v* = ∑ ki vi*	ki = v*(vi)	v = ∑i ki vi
	dot product		

v*(w) = [v]b ● [w]b

v = ∑ v*(vi) vi
= ∑ ([v]b ● [vi]b) vi

***) Matrix representation**

Let b={vi} be a basis of V.
Let v ∈ V.

v*(w)	= [v]b ● [w]b	
	= [v]b [w]b^T	matrix multiplication
v	= ∑i [v]b ● [vi]b vi	
	= ∑i v*(vi) vi	
	= B [v]b	where B=(v1 ... bn) basis matrix
[v]b	= B^-1 v	

Let b' = {v'i} be another basis.
Then

v*(w) in b'	= [v]b' ● [w]b'	
	= [Ev]b ● [Ew]b	where E = B B'^-1
	= Ev*(Ew) in b	

Matrix multiplication is also a representation of the **duality**: dual of **column F^m** vs **row F^n***

Let A = {aij} be an m x n matrix in F and x = {xi} ∈ F^n
Then

A x	= ∑ xj aj	**linear combination** of **column** vectors aj ∈ F^m
	= { Ai*(x) }i	**application** of **row** vectors Ai* ∈ F^n*
	= { x*(Ai) }i	commute
	= { Ai ● x) }i	as dot product

In particular, if **E is orthonormal** i.e. E^-1 = E^T, then

E [v]e	= ∑ v*(vi) vi = v	v as linear combination of basis vi
E^T v	= { v*(vi) }i = [v]b	coefficient v*(vi) of v in basis vi

***) Dual V* vs Hom(V, W)**
https://en.wikipedia.org/wiki/Tensor_product#Linear_maps_as_tensors

Linear functional v*:V → F
Linear map (homomorphism) φ: V → W
Let V, W be vector spaces with **basis vi** and **wj** respectively.
The main observation is that
a) every **linear map φ: V → W** is completely determined by the **values φ(vi)**
on basis vi.
b) every φ(vi) is determined by **basis wj**
i.e.

φ: V → W ∈ Hom(V, W)
φ(vi) = ∑ wj*(φ(vi)) wj ∈ W
= ∑ dij wj dij= wj*(φ(vi)) i.e. coefficient of φ(vi) in basis {wj}

Hence

φ(v) = ∑i vi*(v) φ(vi)

= ∑i vi*(v) ∑j dij wj

= ∑i,j [vi*(v) · dij) wj

Thus, this suggests a 1-1 correspondence
i.e. between **V*⊗W** and **Hom(V,W)** when V is finite

There is a 1-1 mapping between the basis elements

V*⊗W	**<=>**	**Hom(V,W)**
vi*⊗wj		[vi*,wj]: V → W [vi*,wj](vi) = wj [vi*,wj](vi') = 0

note: basis since for any φ: V → W

φ = ∑i [vi,φ(vi)]

i.e.

φ(v) = ∑i [vi,φ(vi)](v)
= ∑i vi*(v) φ(vi)

In particular, any **linear map φ: V → W** corresponds to a vector in V*⊗W

Hom(V, W)	**=>**	**V*⊗W**	
φ: V → W			
φ = ∑i [vi, φ(vi)]		∑i vi⊗φ(vi)	
= ∑i,j [vi, ∑j kjwj]		= ∑i,j kj vi⊗wj	note: kj = wj*(φ(j))

And hence

dim V*⊗W = (dim V)(dim W)
= dim Hom(V,W)

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== Determinant ==

https://en.wikipedia.org/wiki/Determinant

Determinant
The determinant is simply an **alternating form** det: GLn → F on matrices.
i.e.

det: GLn → F

identity	det I = 1		
multilinear	det (....ku ...)	=	k det (...u...)
	det (...u+v...)	=	det (...u...) + det (...v ...)
alternating	det (... v ... v...) = 0		
	Hence anticommutative		
	det (... v....u...) = -det (...u...v...)		

By multilinear expansion, we obtain an **explicit** equation called the **Leibniz** formula.

det A = $\sum$ det (a1,σ(1)eσ(1) a2,σ(2)eσ(2) ... an,σ(n)eσ(n))
by multilinear expansion
$\sigma \in$ Sn
all combination of one value from each column

= $\sum$det(eσ(1) ... eσ(n)) $\prod$i ai,σ(i)
$\sigma \in$ Sn

= $\sum$ **sgn(**σ**)** $\prod$**i ai,**σ**(i)**
Leibniz formula
$\sigma \in$ Sn
by rearrangement det (eσ(1)...eσ(n)) = sgn(σ) det(I)

Equivalently an exterior product:

a1 $\wedge$ **...** $\wedge$ **an** = $\sum$ **sgn(**σ**)** $\prod$ **ai,**σ**(i)** (e1$\wedge$...$\wedge$en)
$\sigma \in$ Sn

= det (A) (e1$\wedge$...$\wedge$en)

Thus, the determinant also has a **recursive** formulation called the **Laplace** formula
For any

det A **= ∑j (-1)^i+j ai,j det Ai,j** for any i
Ai,j = A with row i, column j removed

and hence Cramer's formula

(det A) I **= A (adj A)** where adj A = { (-1)^i+j det Aj,i }

note: **Aj,i** NOT Ai,j

Which provides a formula for A's **inverse**:

A^-1 = 1/det A (adj A)

A is invertible <=> det A ≠ 0

Interpretation: canonical **multiplicative invariant**
The determinant can also be defined as a multiplicative invariant.
i.e. a **multiplicative** function that is **invariant** under choice of **basis**:

invariant	det A	= ∏ λi	λi = ith eigen value
multiplicative	det AB	= det A det B = det B det A	

Put another way, it is a **canonical** analogue of **multiplication** for matrices.

	scalar	**~**	**matrix**
identity	a1 = a		det I = 1
commutative	ab = ba		det AB = det A det B = det BA

Interpretation: solution of a system of linear equations

A^-1 = 1/det adj(A)

Ax = y

x = A^-1 y

Theorem: Determinant definitions
Let A = (aij) be an n x n matrix.

1: Multilinear formulations

1a) Explicit formula

Leibniz formula

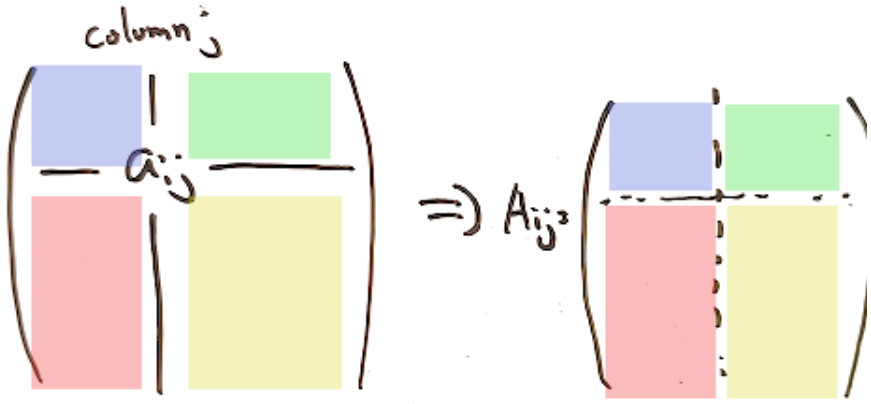
$$\det A = \sum_{\sigma \in Sn} \operatorname{sgn}(\sigma) \prod_i a_{i,\sigma(i)}$$

Equivalently, the **Laplace expansion**

$$\det A = \sum_i a_{i,j} C_{i,j}$$ Cofactor $C_{i,j}$

$$= \sum_i (-1)^{i+j} a_{i,j} \det A_{i,j}$$

for any j.



1b) **Alternating multilinear form**

The determinant is the **unique alternating multilinear form** on square matrices with the properties:

identity	det I	=	1
multilinear	det (...., ku, ...)	=	k det (..., u, ...)
	det (..., u+v, ...)	=	det (..., u, ...) + det (..., v, ...)
alternating	det (... v ... v...)	=	0

note: u, v are **column vectors**

Equivalently, the determinant is a concrete form of the **wedge product** **a1∧...∧an**

$$\mathbf{a1} \wedge \ldots \wedge \mathbf{an} = \sum_{\sigma \in Sn} \operatorname{sgn}(\sigma) \prod \mathbf{a}_{i,\sigma(i)} (e1 \wedge \ldots \wedge en)$$

$$= \det(A)\,(e1 \wedge \ldots \wedge en)$$

2: Multiplicative definitions

2a) **Multiplicative property**

The determinant is a unique function f: GLn → F such that

identity	f(I)	=	1
multiplicative	f(AB)	=	f(A) f(B)

triangular f(A) = ∏ai,i if A is an upper/lower triangular matrix

then f = det.

In particular, the determinant is a **basis invariant**

det (A) = ∏ λi λi = ith **eigenvalue**
since det(A) = det (E^-1AE) E = matrix of eigen vectors (v1 ... vn)
= det (Λ) Λ = diagonal matrix of eigenvalues {λi}
AE = EΛ

since eigenvalues are basis invariant.

2b) **Elementary transformation**
The determinant is the **unique** function on matrices with the following properties for **elementary column operations**

scale	preserve	det (..., ku, ...)	=	k det (..., u, ...)
add	invariant	det (..., u, ..., v, ...)	=	det (..., u, ..., ku+v, ...)
flip	alternate	det (..., u, ..., v, ...)	=	- det (..., v, ..., u, ...)

note: u, v are **column vectors**

note: flip is derivable from scale and add

(u, v)	=> (u, u + v)	+ 0 add
	=> (-u, u+v)	* -1 scale
	=> (-u + u + v, u+v)	+ 0 add
	=> (-v, u+v)	* -1 scale
	=> (-v, u)	+ 0 add
	=> (v, u)	* -1 scale

and

det I = 1

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Proof

Case 1: multilinear formulations
Just apply exterior algebra.
If

Det (e1∧...∧en) = 1

then d (a1∧...∧an) is uniquely defined by

$$\text{Det}(a1\wedge\ldots\wedge an) = \sum_{\sigma \in Sn} \text{sgn}(\sigma) \prod ai,\sigma(i)$$

Hence d is unique + Leibniz.

Laplace expansion:
Pick any j, then

det A = $\sum_i a_{i,j} \left(\sum_{\sigma(i)=j} \operatorname{sgn} \sigma \prod a_{k,\sigma(k)}\right)$

= $\sum_i (-1)^{(j-i)} a_{i,j} \det A_{i,j}$

= $\sum_i (-1)^{(j+i)} a_{i,j} \det A_{i,j}$ since j+i is even <=> j-i is even

Case 2: Multiplicative formulations
Express the column operations as elementary matrices:

	Matrix	determinant
scalar multiple	column j = cej	c
column sum	column j = ek + ej	1
flip	column j = ek column k = ej	-1

Then
* case: multiplicative => transform
The determinant of the elementary matrices are:

scalar multiple by k	det E = c $\prod$ 1 = c	by the diagonal formula
column sum	det E = $\prod$ 1 = 1	by the diagonal formula

Hence

scalar multiple by k	det AE = k det A
column sum	det AE = det A

* case: transform => multiplicative
Decompose A, B into elementary matrices

A = E1....El
B = F1....Fk
AB = E1....El F1....Fk

Then by applying the transforms

det A = det E1 ... det El
det B = det F1 ... det Fk
det AB = det E1 ... det El det F1 ... det Fk

=> det A det B = det AB

Case 3: Multilinear <=> multiplicative

QED
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Interpretation: invariant

Basic properties
Cofactor Ci,j
Minor Mi,j
Adj A
Let A be a matrix.

submatrix Ai,j = A with row i, column j removed

Then

minor Mi,j = det Ai,j

cofactor Ci,j = (-1)^i+j Mi,j

The adjugate adj A is the transpose of cofactors

adj A = { Cj,i }

More generally, for any **row index set I** and **column index set J**

submatrix AI,J = {ai,j} $i \in I, j \in J$ i.e. A with only the I and J columns

minor MI,J = det AI,J

Theorem: Basic properties

Basic

transpose det A = det A^T

since $\sum \text{sgn } \sigma \prod_i a_{i,\sigma(i)}$
$= \sum \text{sgn } \sigma \prod_j a_{\sigma^{-1}(j),j}$ with $j = \sigma(i)$
$= \sum \text{sgn } \sigma \prod_j a_{\sigma(j),j}$ since $\text{sgn } \sigma = \text{sgn } \sigma^{-1}$

multiplicative det AB = det A det B
= det BA

inverse det A^-1 = (det A)^-1
= 1/det A

triangular det A = $\prod$ ai,i if A is triangular

zero det A = 0 if A has a zero row or column

Inverse

Let A by an n x n matrix. Then

A^-1 exists **<=>** **det A ≠ 0**

Cramer's rule **(det A) I** **= A (adj A)**

Or per row i

det A = ∑j ai,j Ci,j

= ∑j (-1)^i+j ai,j (det Ai,j)

Hence

A^-1 **= 1/(det A) adj A**

Gaussian elimination
TODO

Derivative
https://en.wikipedia.org/wiki/Jacobi%27s_formula

For any ai,j

d det A / dai,j = Cij
= (-1)^i+j det Ai,j simply by det A = ∑j ai,j Ci,j

=> d det A/ dai,j = Ci,j

Hence generally, if A = A(t)

d det A / dt = tr (Adj A · dA/dt)

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Proof

*) Inverse

A has an inverse <=> A reduces to row echelon form via matrix row operations
with non-zero diagonals

<=> det A ≠ 0
note: since det A = k det A' for some k ≠ 0, A' = A reduced to row echelon form

Cramer's rule
Let B be an nxn matrix.

Let **Si,j** be permutation **group Sn-1** but the terms labeled differently.

Si,j = permutations σ**: Zn/{i}** $\rightarrow$ **Zn/{j}**

Si,j ≃ Sn-1 isomorphism

Then

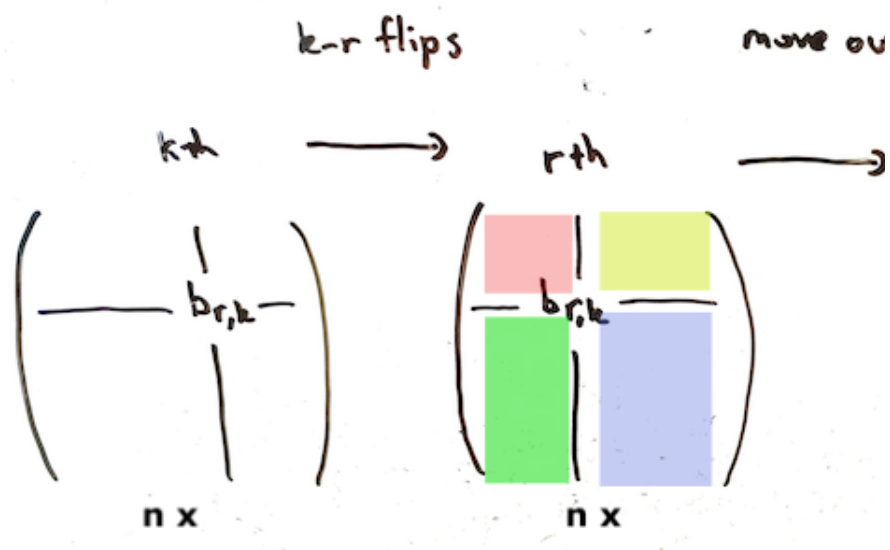
det B = $\sum_{\sigma \in Sn} \text{sgn}\ \sigma \prod_{k} b_{k,\sigma(k)}$

Group by **row r**

$= \sum_{j} \sum_{\sigma \in Sn,\ \sigma(k) \neq j,\ k \neq r} \text{sgn}\ \sigma \prod_{k} b_{k,\sigma(k)}$ Group by **row r**

Switch to **permutations π: Zn/{r} → Zn/{j}**

$= \sum_{j} (-1)^{r-j} \sum_{\pi \in Sr,j} \text{sgn}\ \pi \prod_{k} b_{k,\pi(k)}$ sgn σ = (-1)^r-j sgn π since **r-j flips** to permute r and j



$= \sum_{r} b_{r,j} (-1)^{r+j} B_{r,j}$ note: (-1)^r-j = (-1)^r+j

Hence

$\sum_{j} a_{k,j} (-1)^{i+j} A_{i,j} =$ = det A' where A' = A with **row i** replaced by **row k**

= **det A** if i = k since A' = A

0 o.w. since A' has **duplicate** rows in row i and k

=> A (adj A) = (det A) I

*) Jacobi's formula

d det A(t) / dt = $\sum_{i,j}$ d det A/ dAi,j dAi,j / dt by chain rule

= tr D where D = diagonal matrix with ith diagonal

∑j d

det A/ dAi,j dAi,j / dt

= tr (Adj A ⋅ dA/dt)

QED

==========

Theorem: Vandermonde matrix
https://en.wikipedia.org/wiki/Vandermonde_matrix

Let

$$V = V(x_0, x_1, \cdots, x_m) = \begin{pmatrix} 1 & x_0 & x_0^2 & \dots & x_0^n \\ 1 & x_1 & x_1^2 & \dots & x_1^n \\ 1 & x_2 & x_2^2 & \dots & x_2^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_m & x_m^2 & \dots & x_m^n \end{pmatrix}$$

If n = m, then

det V = $\prod_{i<j}$ (xj - xi)

= (-1)^N $\prod_{i<j}$ (xj - xi) where N = n(n+1)/2 = n+1 choose 2

==========

Proof
By the Leibniz form

det V = $\sum_{\sigma \in Sn}$ sgn σ $\prod$ xi^σ(i)

=> det V is a polynomial in x0, ..., xn and each term has degree

1 + ... + n = n(n+1)/2

claim: xi - xj divides V
If xj = xi, then row i and j are equal. Hence

det V[xi = xj] = 0

=> xi = xj is a solution of det V
i.e. xi - xj divides V

QED claim

Hence

$\prod_{i<j}$ (xj - xi) | det V

On the other hand, there are

n+1 choose 2 = n(n+1)/2

possible combinations of xi - xj
i.e. each term of the product is of degree n(n-1)/2

=> the division is exact

QED
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== Trace ==

https://en.wikipedia.org/wiki/Trace_(linear_algebra)

Trace

Theorem: Equivalent formulations

a) **Diagonal sum**
Trace is the sum of the diagonals

tr(A) **= ∑i aii**

= ∑ ei ● Ai Ai = ith column

b) **Additive invariant**
Trace is the sum of the eigenvalues

tr(A) **= ∑i λi**

c) **Linear form**
Trace is the unique linear form

identity	tr(I)	= n
linear	tr(A+B)	= tr(A) + tr(B)
	tr(cA)	= c tr(A)
semi-commute	tr (AB)	= tr(BA)

==========
Proof:
case: diagonal => linear

tr(AB)	= ∑i (∑k aik bki)	ith row of A ● ith column of B
	= ∑k (∑i bki aik)	kth row of B ● kth column of A
	= tr (BA)	

case: linear => diagonal
Let

Xi,j = matrix with 1 in (i,j), 0 otherwise

It suffices that

tr Xi,j = δi,j
i.e. 1 if i=j, 0 otherwise

Since then

tr A = $\sum$ aij tr Xi,j
= $\sum$ aii

claim: tr Xi,j = δi,j
For any i

Xi,i = Xj,i Xi,j for any j

=> tr Xi,i = tr Xj,i Xi,j
= tr Xi,j Xj,i
= tr Xj,j

=> n = tr I
= $\sum$ tr Xi,i
= n x where x = tr Xi,i for any i

=> tr Xi,i = 1

On the other hand, for any i ≠ j

Xi,j = Xi,k Xk,j

=> tr Xi,j = tr Xi,k Xk,j
= tr Xk,j Xi,k
= tr 0
= 0

QED claim

case: invariant => linear
Trivial

case: linear => invariant
Let

AE = EΛ where Λ = {λi} diagonal matrix of eigenvalues
E = (v1 ... vn) matrix of eigenvectors

be the decomposition of A by its eigenvalues.
Then

tr(A) = tr (AEE^-1)
= tr(E^-1AE)
= tr(Λ)
= $\sum$ λi

QED
==========

Theorem: Basic properties

Basic

Vector space trace encodes the **dimension** of the vector space

tr I = n
= dim V

Invariant tr A = $\sum \lambda i$
= $\sum$ aii

multiplicative tr(AB) = tr(BA)

Hence:
Circular shift

tr(ABC) = tr(CAB)
= tr(BAC)

Invariant under change of basis E

tr(A) = tr(E^-1 A E)

transpose tr(A) = tr(A^T)

hence

tr(A B^T) = tr(B^T A)
= tr(A^T B) by transpose
= tr(B A^T)
= $\sum$ aij bij

inverse tr(A^-1) = $\sum 1/\lambda i$

Hence, in general, there is no simple relation between tr(A) and tr(A^-1)

A special case is if λi is a **root of unity**
i.e. $|\lambda i| = 1$ for all i for complex $\lambda i \in C$

tr(A^-1) = $\sum 1/\lambda i$
= $\sum \lambda i^{-}$ complex conjugate
= **tr(A)$^{-}$**

characteristic polynomial The characteristic polynomial of A is of the form

det (A - λI) = $\sum$ sgn $\sigma \prod$ (ai,σ(i)-λI)

= (-λ)^n + (-λ)^n-1 $\sum$aii + ...

Hence

coefficient of λ^n-1 = (-1)^n-1 tr A

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== Direct Sum, Tensor ==

Direct sum V⊕W

Let V, W be vector spaces
Define V⊕W by

vector	v⊕w ∈ V⊕W	
sum	**v⊕w + v'⊕w'**	= v⊕w' + v'⊕w **= (v+v') ⊕ (w+w')**
scale	**c (v⊕w)**	**= cv⊕cw**

V⊕W is a vector space over field F:

zero	0⊕0	
commutative	v⊕w + v'⊕w'	= (v+v') ⊕ (w+w')
by sum definition		= (v'+v) ⊕ (w'+w)
by commutativity of V and W		= v'⊕w' + v⊕w
associative	v⊕w + (v'⊕w' + v''⊕w'')	= (v+(v'+v'')) ⊕ (w+(w'+w''))
by sum definition		= ((v+(v'))+v'') ⊕ ((w+w')+w'')
by associativity of V and W		= (v⊕w + (v'⊕w')) + v''⊕w''

If f: V → R and g: W → S are linear maps, let

f⊕g: V⊕W → R⊕S
f⊕g(v⊕w) = f(v)⊕g(w)

f⊕g is a **linear map**:

sum	f⊕g(v⊕w + v'⊕w')	= f⊕g(v+v'⊕w+w') = f(v+v')⊕g(w+w') = f(v)+f(v')⊕g(w)+g(w') = f(v)⊕g(w)+f(v')⊕g(w')
scale	f⊕g(c (v⊕w'))	= f⊕g(cv⊕cw') = f(cv)⊕g(cw) = cf(v)⊕cg(w) = c (f(v)⊕g(w))

Hence

$$A \oplus B = \begin{matrix} A & 0 \\ 0 & B \end{matrix}$$

tr (A⊕B) = tr(A) + tr(B)

Tensor V⊗W

Let V, W be vector spaces with bases e={vi} and d={wj} respectively.
Define V⊗W

vector	basis vi⊗wj

Then

sum	(vi+vi')⊗(wj+wj')	= vi⊗wj + vi⊗wj' + vi'⊗wj + vi⊗wj'
	hence	
	(vi+vi')⊗wj	= vi⊗wj + vi'⊗wj
	vi⊗(wj+wj')	= vi⊗wj + vi⊗wj'
scale	c v⊗w	= cv⊗cw

Hence, for any $v \in V$, $w \in W$

$v \otimes w = \sum vi^*(v)\, vi \otimes \sum wj^*(w)\, wj$

$= \sum (vi^*(v)\, wj^*(w))\ vi \otimes wj$

V⊗W is a **vector space** over F

If f: V → R, g: W → T are linear maps, let

f⊗g: V⊗W → R⊗T

f⊗g(vi⊗wj) = f(vi)⊗g(wj)

f⊗g(vi⊗wj + vi'⊗wj') = f⊗g(vi⊗wj) + f⊗g(v'j⊗w'j)

Then **f⊗g** is a **linear map**:

linear	by definition	
scale	f⊗g(c v⊗w)	= f(cv)⊗g(cw) = cf(v)⊗cg(w) = c f⊗g(v⊗w)

For simplicity, assume the bases are elementary bases ei = (....1...) in ith position

If A, B are **matrices** of f, g

i.e. $f(v) = Av = \sum (Ai \bullet v)\, ei$ i=1...m Ai = ith row of A

$g(w) = Bw = \sum (Bs \bullet w)\, es$ s = 1...m' Bs = sth row of B

then

f⊗g(v⊗w) = f(v)⊗g(w)

$= \sum_{i,s} (Ai \bullet v)ei \otimes (Bs \bullet w)\, es$

$= \sum_{i,s} (Ai \bullet v)(Bs \bullet w)\ ei \otimes es$

$= \sum_{i,s} (\sum_{j,t} aij\ bst\ vj\ wt)\ ei \otimes es$

As a matrix

A⊗B (v⊗w) = { (Ai • v) (Bs • w) }

$$A \otimes B (v \otimes w) = \begin{pmatrix} A_1 \cdot v & B_1 \cdot w \\ A_1 \cdot v & B_2 \cdot w \\ \vdots & \\ A_n \cdot v & B_{n'} \cdot w \end{pmatrix} \begin{matrix} e_1 \otimes e_1 \\ e_1 \otimes e_2 \\ \vdots \\ e_n \otimes e_{n'} \end{matrix}$$

A⊗B can be seen as a **mm' x nn' matrix**

A⊗B **= { aij B }** i=1...m, j=1...n

= { aij bst } i=1...m j=1...n
s=1....m' t=1...n'

$$A \otimes B = \begin{pmatrix} a_{11} B & \cdots & a_{1n} B \\ & \ddots & \vdots \\ a_{m1} B & \cdots & a_{mn} B \end{pmatrix}$$

$mm' \times nn'$

with vectors v = ∑ ci ei, w = ∑ ds es

$$v \otimes w = \begin{pmatrix} c_1 d_1 & e_1 \otimes e_1 \\ c_1 d_2 & e_1 \otimes e_2 \\ \vdots & \\ c_1 d_{n'} & e_1 \otimes e_{n'} \\ c_2 d_1 & \\ \vdots & \\ c_n d_{n'} & e_n \times e_{n'} \end{pmatrix}$$

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== OLD ==

f⊗g(ej⊗et) = f(ej)⊗g(et)

= **∑i (Ai●ej) ei ⊗ ∑s (Bs●et) es**

= ∑ (aij bst) ei⊗es
i,s

Hence for any v,w

f⊗w(v⊗w) = ∑ ej*(v) et*(w) f⊗w(ej⊗et)
j,t

= **∑ (∑ ej*(v) et*(w) aij bst) ei⊗es** by rearrangement
i,s j,t

= ∑ (Ai ● v)(Bs ● w) ei⊗es
i,s

= ∑ (Ai ● v)ei ⊗ (Bs ● w) es
i,s

Hence A⊗B can be seen as a **mm' x nn' matrix**

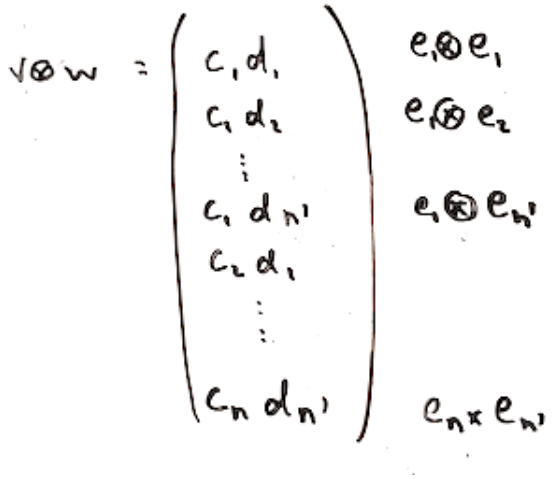
A⊗B **= { aij B }** i=1...m, j=1...n

= { aij bst }

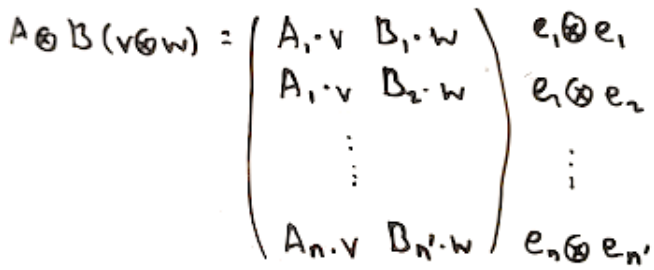
$$A \otimes B = \begin{pmatrix} a_{11}B & \cdots & a_{1n}B \\ \vdots & \ddots & \vdots \\ a_{m1}B & \cdots & a_{mn}B \end{pmatrix}$$

$mm' \times nn'$

with vectors v = ∑ ci ei, w = ∑ ds es



A⊗B (v⊗w) = { (Ai • v) (Bs • w) }



Hence if R=V and T=W, then trace

tr(A⊗B) = ∑ aii bss

= tr(A) tr(B)

***** TODO *****

V*⊗W		**Hom(V,W)**	
ei*⊗dj	<=>	ei* dj: V → W ei*(ei) = dj ei*(ek) = 0 for i ≠ k	**basis dj** of W
ei*⊗w	<=>	ei* w: V → W	any **w = ∑i ki dj** ∈ **W**
= ∑j kj (ei*⊗dj)		= ∑j kj (ei* dj)	

Hence, summing over all ei

∑i ei*⊗wi	<=>	∑i ei* wi
= ∑i,j ki,j ei*⊗dj		= ∑i,j ki,j ei* dj

In particular, for any φ: V → W:

Hom(V, W)		**V*⊗W**
φ: V → W	<=>	
φ = ∑i ei* φ(ei)		∑i ei*⊗φ(ei)

NOTE: only if sum is finite

components

ei* φ(ei) <=> ei*⊗φ(ei)

Conversely, for any f⊗w ∈ V*⊗W

V*⊗W		**Hom(V,W)**
f: V → F ∈ V*	=>	φ: V → W
f⊗w		φ
= ∑ ei* f(ei) ⊗ w		= ∑i **ei*** **f(ei)w**
= ∑i **ei*⊗ f(ei) w**		
NOTE: sum interchange only if sum is **finite**		

components

ei* ⊗ f(ei)w <=> ei* f(ei)w

Note though that the ∑i ei*⊗· exists only if V is finite dimensional.
Thus, if V is finite dimensional, then V*⊗W and Hom(V,W) are isomorphic

isomorphism Hom(V,W) → V*⊗W
φ: V → W
φ = ∑i ei* φ(ei) <=> ∑i ei*⊗φ(ei)

dim V*⊗W = (dim V)(dim W)
= dim Hom(V,W)

Det: ∧V → F

where

Det (e1∧...∧en) = 1 i.e. **det I = 1**

This is sufficient to **uniquely** define the determinant, since for any vectors ai = ∑ ai,j ej

Det (a1 ∧ ... ∧ an) = ∑ sgn(σ) ∏ ai,σ(i) since a1 ∧ ... ∧ an = ∑ sgn(σ) ∏ ai,σ(i) (e1∧...∧en)
σ ∈ Sn

i.e. Leibniz formula.
Write

det (a1, ..., an) = Det (a1 ∧ ... ∧ an)

trace: linear form

invariant	**tr A**	**= ∑ λi**
linear	tr (A + B)	= tr(A) + tr(B)
	tr (cA)	= c tr(A)
	tr (AB)	= tr(BA)

determinant: alternating form

invariant **det A** **= ∏ λi**

alternating form
det (...a + b...) = det (...a...) + det (...b...)
det (....ca...) = c det(...a...)
det (...a...a...) = 0

hence

det (...a...b...) = - det (...b...a...) anti-commutative

det AB = det A det B multiplicative
= det BA

as wedge product

a1∧...∧an = det A e1∧...∧en
where A = (a1...an)

The trace and determinant can be viewed as the simplest forms

trace

tr(A+B) = tr(A) + tr(B) ∑(λi + ηi) = ∑λi + ∑ηi sum

tr(cA) = ctr(A) ∑ cλi = c∑λi scalar

tr(AB) = tr(BA) ∑λi ηi = ∑ηi λi

? ∑λi ∑ηj = ∑i,j λj ηj

determinant

det AB = det A det B ∏ λiηi = ∏λi ∏ ηi

? ∏ (λi+λ'i) = ∏ ηi where ηi = λi or λ'i

Let A' be A with **row x** copied into **row y**

det A' = ∑ sgn σ ∏l a'l,σ(l)

Group by products on **row y**
(-1)^y-k since y-k flips to move from (y,k) position to (y,y)

= ∑ ax,k (-1)^y-k [∑ sgn σ ∏ al,σ(l) where l ≠ y, σ(l) ≠ k]
k l

= ∑ ax,k (-1)^y+k Ay,k note: (-1)^k-y = (-1)^k+~!
k

= ∑ ax,k Ck,y

k

Hence

k = i det A' = det A since A' = A

k ≠ i det A' = 0 since A' = A with row i = row k

=> (det A) I = A (adj A)

B'

∑ ai,k (-1)^j+k Bj,k = ∑ ai,k (-1)^j+k ∑ sgn π ∏l bl,π(l)
where l ≠ i, π(l) ≠ l

= ∑ sgn σ ai,k #

∑ ai,k Ck,j = ∑ ai,k (-1)^j+k Aj,k

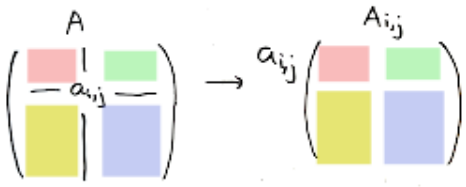
where Aj,k = ∑ sgn π ∏ al,π(l) where
row j, column k are skipped

π ∈ Sn-1

∑ sgn(σ) ∏ ak,σ(k) = ∑ ai,j (∑ sgn σ ∏ ak,σ(k))
σ ∈ Sn k σ(i) = j

= ∑ ai,j (-1)^j+i (∑ sgn π ∏ ak,π(k)) j-i
column flips to move ai,j in cell (i, j) to (i, i)

π ∈ Sn-1



= ∑ (-1)^j+i ai,j det Aij note:
(-1)^j-i = (-1)^j+i

= ∑ ai,j Ci,j

= cell i,i of A(adj A)

On the other hand,

cell i,k of A(adj A) = ∑ ai,j Ck,j

= ∑ (-1)^j+i ai,j det Akj

= 0 since two Ak,i

∑j aij Ci,j = ∑j aij (-1)^i+j det Ai,j

= ∑ sgn(σ) ∏ ai,σ(i) since

	=	det A
∑j aij Ck,j	=	∑j aij (-1)^k+j det Ak,j
	=	0
=> A adj(A)	=	(det A) I

If V has a dot product, then v* is just the dot product with v

coefficients v*: V → K

v*(w) = v*(∑i w*(ei) ei)
= ∑i w*(ei) v*(ei)

= [v] ● [w] [v] = vector (v*(e1) v*(e2) ...)

= [v] [w]^T matrix multiplication

vector v ∈ V

v = ∑i v*(ei) ei

= ∑i (v●ei) ei

= E [v] where E = (e1 ... en)

Thus, for any f⊗w ∈ V*⊗W:

V*⊗W		**Hom(V,W)**
f: V → K ∈ V*	=>	φ: V → W
f⊗w		φ
= ∑ ei* f(ei) ⊗ w		= ∑i **ei* f(ei)w**
= ∑i **ei*⊗ f(ei) w**		
NOTE: sum interchange only if sum is **finite**		
components		
ei* ⊗ f(ei)w	<=>	ei* f(ei)w

Conversely, for any φ: V → W ∈ Hom(V,W):

Hom(V, W)		**V*⊗W**
φ: V → W	<=>	
φ = ∑i ei* φ(ei)		∑i ei*⊗φ(ei) NOTE: only if sum is finite
components		
ei* φ(ei)	<=>	ei*⊗φ(ei)

In general, for any w ∈ W, there is a 1-1 correspondence between individual **components ei*⊗w**:

	V*⊗W		**Hom(V,W)**
basis dj of W	ei*⊗dj	<=>	ei* dj: V → W

any $w = \sum_j k_j d_j \in W$	$e_i^* \otimes w$	<=>	$e_i^* w: V \to W$
	$= \sum_j k_j (e_i \otimes d_j)$		$= \sum_j k_j (e_i^* d_j)$

Generally

$\sum_i e_i^* \otimes w_i$	<=>	$\sum_i e_i^* w_i$
$= \sum_{i,j} k_{i,j} e_i^* \otimes d_j$		$= \sum_{i,j} k_{i,j} e_i^* d_j$

where $w_i = \sum k_{i,j} d_j$ for each i

If V (and hence V*) is **finite dimensional**, then sums of the form $\sum_i e_i \otimes \cdot$ always exist.
Hence, $V^* \otimes W$ and Hom(V,W) are **isomorphic**

isomorphism Hom(V,W) → $V^* \otimes W$
$\varphi: V \to W$

$\varphi = \sum_i e_i^* \varphi(e_i)$	<=>	$\sum_i e_i^* \otimes \varphi(e_i)$
		NOTE: sum must be finite

dim $V^* \otimes W$ = (dim V)(dim W)
= dim Hom(V,W)

If $f \in V^*$, $w \in W$ then φ defined by

$\varphi: V \to W$
$\varphi(v) = \varphi(v)w$

linear

zero	$\varphi(0)$	$= f(0)w$
		$= 0$
scalar multiple	$\varphi(kv)$	$= f(kv)w$
		$= k F(v)$
addition	$\varphi(v+v')$	$= f(v+v')w$
		$= f(v)w + f(v')w$
		$= \varphi(v) + \varphi(v')$

is a **homomorphism** V → W i.e. linear map.

$f \otimes w: V \to W$
$f \otimes w(v) = f(v)\, w$

tensor

linear	$(f + g) \otimes w\,(v)$	$= (f + g)(v)\, w$
		$= f(v)w + g(v)w$
		$= f \otimes w\,(v) + g \otimes w\,(v)$
	$f \otimes (w + w')\, v$	$= f(v)(w + w')$
		$= f(v)w + f(v)w'$
		$= f \otimes w + f$

Conversely,

φ: V → W
φ(v) = ∑i ki φ(ei)

φ(ei) = ∑j lj fj

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